

Contents

The Problem: Bots Don't Look Like Bots Anymore

A CREDENTIAL-STUFFING CAMPAIGN LOOKS IDENTICAL TO HUMAN LOGIN TRAFFIC at the HTTP layer. The requests are syntactically valid. The user-agent strings are plausible. Your CDN's bot-score misses it because the attacker rotates IP addresses faster than reputation feeds update.

1

Traditional TLS-aware defences — SSL inspection and decryption — introduce significant latency and create a massive compliance exposure: PCI-DSS, HIPAA, and GDPR all restrict access to decrypted traffic. JA4proxy solves this by enforcing security *before* decryption, operating directly on the ClientHello handshake.

¹ The Verizon DBIR consistently shows that credential abuse accounts for more than 40% of breaches. Most credential-stuffing toolkits ship with TLS stacks that produce distinctive, stable fingerprints — yet traditional defences never inspect the handshake.

The JA4 Insight

BEFORE A SINGLE BYTE OF ENCRYPTED DATA IS EXCHANGED, every TLS client announces its capabilities in cleartext. This announcement — the ClientHello — contains the TLS version, cipher suites, and extensions (SNI, ALPN).

Because this information reflects the client's TLS stack, it is remarkably stable. Sliver 1.5.x produces the same ClientHello whether it is running from Frankfurt or Tokyo. Upgrading the *payload* has no effect on the fingerprint.

2

Client	JA4 Fingerprint (example)	Disposition
Chrome 125 (Win)	t13d1516h2_8daaf6152771_02713d6af86	ALLOW
Firefox 126	t13d1915h2_cd87aab0d655_5d1a4de8bd1	ALLOW
Sliver C2	t13d2012_c1e4e3a9b26a_000000000000	BLOCK
Cobalt Strike	t13d1112_b4f47bba57b5_9b5e0e9c9d74	BLOCK

² **Elite Performance:** The JA4proxy Go core processes these fingerprints in less than 27 microseconds, enabling theoretical throughput of 6 million connections per second per core. C2 frameworks typically do not.

Architecture: The Ultra-Lite Go Engine

JA4PROXY V2.0.0 IS A PURE GO-NATIVE IMPLEMENTATION designed for extreme resource efficiency and sub-microsecond latency. The system is distributed as exactly two binaries:

- **JA4proxypd (The Engine):** A stateless, high-concurrency traffic daemon that handles TLS-passthrough at scale.
- **JA4proxyp (The Toolset):** A unified operational CLI for management, real-time simulation, and integrated high-speed benchmarking.

High-Performance Observability

OPS TEAMS REQUIRE MORE THAN *average* LATENCY METRICS. JA4proxy surfaces p99 tail latency in real-time, ensuring that security enforcement never causes intermittent jitters for legitimate users.

Traffic Profile	p99 Latency	CPU Impact
Whitelisted (Fast Path)	272 ns	< 1%
Scored (Standard Path)	2.3 ȳ3bcs	< 2%
Adversarial (JA4X)	6.0 ȳ3bcs	< 5%

³JA4proxypd has a memory footprint of less than **12 MB** and starts in under **50 ms**, making it ideal for containerised auto-scaling environments.

Table 2: Go-native performance matrix. Measurements taken on commodity server hardware (Intel i9-9900K).

Ready for Enterprise

JA4PROXY V2.0.0 IS PRODUCTION-CERTIFIED for high-traffic DMZ deployments. The system includes integrated cluster synchronization, Zero-Trust Redis hardening (TLS), and PROXY protocol v2 support with TLV-based trust verification.

Source code and technical evaluators guide:
<https://github.com/seanpor/JA4proxy>
JA4 Specification: <https://github.com/FoxIO-LLC/ja4>

